

Arjun Mittal

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Education

University of Toronto | BAsC in Computer Engineering

September 2016 – April 2021

Coursework - Machine Learning, Algorithms & Data Structures, Operating/Distributed Systems, Computer Networks

Technical Skills

Languages | Python • C++ • C • JavaScript • Go • Java • SQL • MongoDB • KDB

Tooling & Frameworks | React.js • Express.js • Node.js • Django • TensorFlow • PyTorch • Matplotlib

Development Tools & Environments | Git • Bash • Docker • GCP • AWS • Jira • Figma

Experience

RBC Capital Markets | ETF – Algorithmic Trading Software Engineer

January 2020 – August 2020

- Developed trading interface using **React JS** and **Java**, allowing **concurrent traders** to make **real-time system** updates
- Designed and managed a back-testing service using **Python** with **NumPy/Scikit-learn** for data cleaning & normalization
- Reduced run time of scripts by **80%**, improving availability of critical reports through **vectorizing query operations** and **refactoring codebase** from **R** to **KDB**

RBC Capital Markets | ETF – Quantitative Trading Analyst

May 2019 – December 2019

- Implemented an **ETL pipeline** using **KDB**, **SQL**, and **Python** to track profit metrics, decreasing query time by **100x**
- Led the development of an ETF validation tool, using **React JS** and **Python**, resulting in an annual savings of **\$65,000**
- Created **MapReduce** equivalent jobs to optimize **2000 system parameters** with over **1 million** rows of pricing data, increasing ETF market share by **8%** within 2 months

aUToronto | Autonomous Car - Simulation Developer

July 2018 – April 2019

- Interfaced **ROS & CARLA** to create a simulation environment to test autonomous driving algorithms
- Constructed **3D models** of traffic signs to be used in training by perception sub-team

Scotiabank | Payments - Software Engineering Intern

May 2018 – August 2018

- Engineered a **Big Data pipeline** to actively parse and index millions of transactions into **Google Cloud SQL (GCP)**
- Architected a **real-time dashboard** for senior executives using **Python** and **Node JS** to monitor customer KPI's

Projects

Facebook Libra – Smart-Contract Verification | C, React JS, JavaScript, AWS, Docker

2020

Developed a system for the **automatic verification** and **synthesis** of Facebook's Libra smart-contracts

CockroachDB – Open-Source | Go

2020

Contributed to **CockroachDB**, a **distributed SQL database**, expanding its geospatial capabilities

AirBnB Predictor | Python – PyTorch, Pandas

2019

Built a fully connected **RNN network** which estimates listing price of a house based on its feature list

Wealthsimple InvestIt | JavaScript – Express JS, Node JS, MongoDB

2018

Created an equity crowdfunding app that reinvests capital generated from users, using **Wealthsimple API**

Community & Awards

L3 Wescam Drone Hackathon Winner | Placed in the top 3 out of 25 teams in drone hackathon

2019

VEEP | Developed software for NPOs supporting mental health education

2018

Waterloo EngHack - Category Finalist | Awarded best food industry hack out of 32 teams

2018